POWER PULSE

AC and DC Bridges Important Formulae

Notes by Balaji Sir

1 General balance

1. $$Z_1 Z_4 = Z_2 Z_3 \quad \text{(product of opposite arms)}$$

for both the magnitude and phase conditions

2 DC bridges

2. $$\text{Wheatstone: } R_x = \frac{R_2}{R_1} R_3 \quad \text{(medium resistance } 1\,\Omega\text{–}0.1\,M\Omega\text{)}$$

balance gives zero galvanometer current

3. $$\text{Kelvin double bridge: } R_x = S\,\frac{Q}{P} \quad \text{when } \frac{P}{Q} = \frac{p}{q}$$

for low resistance; S = standard resistance

3 Inductance bridges

4. $$\text{Maxwell (L–C): } L_1 = R_2 R_3 C_4, \quad R_1 = \frac{R_2 R_3}{R_4}, \quad Q = \omega C_4 R_4$$

for a low-Q inductor; C_4 in parallel with R_4

5. $$\text{Hay: } L_1 = \frac{R_2 R_3 C_4}{1 + \omega^2 C_4^2 R_4^2}, \quad R_1 = \frac{\omega^2 C_4^2 R_2 R_3 R_4}{1 + \omega^2 C_4^2 R_4^2}, \quad Q = \frac{1}{\omega C_4 R_4}$$

for a high-Q inductor; C_4 in series with R_4

4 Capacitance bridge

6. $$\text{Schering: } C_1 = C_2 \frac{R_4}{R_3}, \quad R_1 = R_3 \frac{C_4}{C_2}, \quad \tan\delta = \omega C_1 R_1 = \omega C_4 R_4$$

used for insulation testing and measuring dielectric loss

5 Frequency bridge

7. $$\text{Wien: } f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \left(= \frac{1}{2\pi RC} \text{ if } R_1 = R_2, C_1 = C_2 \right), \quad \frac{R_3}{R_4} = 2 \text{ (equal values)}$$

used to measure frequency, and as an oscillator element